

CLEVER FIELD, OH, USA

WMO: 725224

Lat: 40.472N

Lon: 81.424W

Elev: 273

StdP: 98.09

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.9	-20.5	0.6	-14.6	-17.9	0.8	-11.8	8.6	1.3	7.9	-0.4	1.4	300	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.1	22.8	30.8	22.3	29.3	21.6	24.5	29.5	23.7	28.3	22.9	27.3	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	27.1	22.3	17.5	26.3	21.5	16.7	25.3	75.3	29.6	71.9	28.3	68.8	27.3	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.6	5.6	-20.1	33.3	3.6	1.7	-22.7	34.6	-24.9	35.6	-26.9	36.5	-29.5	37.8		
(5)			-20.4	26.0	3.5	0.9	-23.0	26.7	-25.0	27.2	-27.0	27.7	-29.6	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-2.0	-0.6	4.5	11.0	16.5	20.8	22.7	21.8	18.4	12.0	6.0	0.9
	DBStd	10.00	6.86	6.29	6.02	5.09	4.49	3.37	2.87	2.76	3.91	4.94	5.16	5.60
	HDD10.0	1384	376	301	193	48	4	0	0	0	1	35	140	287
	HDD18.3	3088	631	529	430	224	91	16	2	4	46	205	369	541
	CDD10.0	1771	3	4	22	78	206	324	393	366	253	97	21	4
	CDD18.3	434	0	0	1	4	33	90	137	112	48	9	0	0
	CDH23.3	3843	0	1	8	74	351	791	1191	908	441	76	2	0
	CDH26.7	1160	0	0	1	9	85	239	407	267	135	17	0	0
Wind	WSAvg	2.2	2.8	2.7	2.8	2.8	2.2	1.9	1.6	1.5	1.6	2.0	2.4	2.6
Precipitation	PrecAvg	1036	75	50	74	89	113	108	111	98	90	72	78	77
	PrecMax	1509	173	120	179	131	206	215	202	201	189	117	137	202
	PrecMin	792	32	21	31	46	50	29	51	29	38	19	28	33
	PrecStd	186	35	24	33	25	42	52	40	44	42	29	27	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	19.3	24.0	27.9	31.2	32.5	33.9	32.8	32.5	29.2	22.3	17.9
		MCWB	13.3	13.8	15.1	17.6	21.0	22.2	23.6	22.9	22.4	21.1	15.9	13.9
	2%	DB	14.6	15.7	20.9	25.7	28.9	31.0	32.1	31.2	30.5	25.3	19.1	14.5
		MCWB	11.7	11.4	13.7	16.1	19.9	22.0	23.4	22.8	21.6	18.3	13.8	11.2
	5%	DB	11.4	12.4	17.6	23.1	27.3	29.3	30.6	29.5	27.9	22.8	17.2	12.3
		MCWB	8.8	9.0	11.8	14.8	19.4	21.4	22.6	22.2	20.6	17.1	12.6	9.8
	10%	DB	7.8	9.0	14.6	20.8	25.3	27.8	29.0	27.9	25.9	20.5	14.8	9.4
		MCWB	5.3	5.8	10.0	13.5	18.2	20.9	21.9	21.4	19.4	16.0	10.9	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	14.7	16.2	19.6	22.8	24.4	25.8	25.3	24.2	21.9	17.4	14.8
		MCDB	16.9	18.4	22.8	25.5	29.0	29.5	31.2	30.4	29.9	27.9	20.7	17.1
	2%	WB	12.3	12.2	14.5	17.4	21.4	23.5	24.8	24.2	22.9	20.1	15.2	12.2
		MCDB	14.6	14.8	19.3	23.0	26.9	28.6	29.9	28.7	27.7	23.7	18.2	13.9
	5%	WB	9.1	9.1	12.9	16.1	20.4	22.7	23.8	23.3	21.8	18.2	13.2	10.0
		MCDB	10.9	12.2	16.6	21.2	25.5	27.7	28.4	27.4	25.9	21.1	16.2	12.1
	10%	WB	5.4	5.8	10.6	14.8	19.2	21.8	23.0	22.4	20.8	16.5	11.4	7.3
		MCDB	7.8	8.9	14.0	19.5	23.9	26.4	27.1	26.3	24.2	19.9	14.5	9.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.3	10.7	12.5	12.2	11.4	11.1	11.3	12.1	11.4	10.1	7.9
		MCDBR	11.3	13.5	15.4	16.3	14.7	13.2	12.5	12.9	14.3	14.5	13.8	11.5
		MCWBR	9.3	10.1	9.4	8.9	7.1	6.1	5.2	5.7	6.5	8.1	9.3	9.6
	5% WB	MCDBR	10.6	12.7	13.5	13.3	12.3	11.1	10.6	10.3	11.3	11.9	11.4	10.8
	MCWBR	9.5	10.4	9.0	8.5	6.5	5.6	5.1	5.3	5.6	7.1	8.8	9.8	
Clear-Sky Solar Irradiance	taub	0.306	0.326	0.358	0.391	0.432	0.459	0.462	0.454	0.393	0.364	0.321	0.306	
	taud	2.441	2.364	2.345	2.280	2.190	2.152	2.163	2.174	2.320	2.397	2.492	2.473	
	Ebn at Noon	856	885	892	881	849	823	817	812	844	830	832	826	
	Edh at Noon	79	99	113	129	144	150	147	142	115	95	75	72	
All-Sky Solar Radiation	RadAvg	1.51	2.37	3.34	4.53	5.37	5.83	5.79	5.32	4.32	2.89	1.93	1.30	
	RadStd	0.13	0.32	0.33	0.45	0.44	0.45	0.41	0.28	0.39	0.26	0.27	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+206	+114
(47) Regional (42 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+68

Nomenclature: See separate page